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Coded aperture temporal compressive digital holographic microscopy

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We report a coded aperture temporal compressive digital holographic microscopy (CATCHY) system to capture high-speed high-resolution samples by integrating snapshot compressive imaging (SCI) into digital holographic microscopy. Specifically, a two-dimensional (2D) detector samples a 4D (x, y, z, t) spatiotemporal data in a compressive manner, and after this, an efficient deep learning-based video SCI reconstruction algorithm is employed to reconstruct the desired 4D data cube. Up to ten high-resolution microscopic images are reconstructed from a snapshot measurement captured by our CATCHY system. Experimental results demonstrate the potential to visualize the 3D dynamic process of micro-nanostructures and imaging biological samples with high spatial and temporal resolution.

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Digital holographic microscopy (DHM) [1–4] is a promising technique to simultaneously capture the amplitude (or intensity) and phase distributions of a target scene. Specifically, DHM utilizes interference to record high-resolution three-dimensional information of the target scene in holograms. To capture high dynamic objects effectively, current DHM systems [5–7] require substantial data accumulation, leading to a subsequent surge in data bandwidth demand. As a result, there is an increasing need to effectively manage and store a large volume of data required for reconstructing high-quality holograms. Therefore, it is desirable to build a new holographic microscopy system to *capture high dynamic objects with low data bandwidth*. Towards this end, this paper proposes a coded aperture temporal compressive digital holographic microscopy system, dubbed as CATCHY, to capture high-speed high-resolution samples by integrating coded aperture compressive temporal imaging [8–10] into DHM. In particular, a two-dimensional (2D) detector samples a 4D (x, y, z, t) spatiotemporal data cube in a compressive manner, and after this, an efficient deep learning-based video SCI reconstruction algorithm is employed to reconstruct the desired 4D data cube. Although a compressive holographic video [11] has been previously reported to allow 3D tomography at different depths from a single 2D compressed image, it missed the phase information. By contrast, our work aims to not only obtain the intensity information but also acquire the phase of the optical field, meanwhile utilizing deep learning-based algorithms to enhance the quality of the reconstructed intensity and phase images. Experimental results show that a compression ratio up to ten can be achieved, demonstrating the effectiveness of our proposed method, and potentially advancing DHM to visualize the dynamic process of micro-nanostructures [6,12] and imaging biological samples [13–15] with high spatial and temporal resolution. In the system design, we adopt the video snapshot compressive imaging (SCI) technology [10,16] to capture dynamic holograms. Compared with the GAP-TV [17] and PnP-FFDNet [18] (see Supplement 1 Fig.S2), we utilize EfficientSCI [19] as the video SCI reconstruction network. Formally, we define a system consisting of a hardware encoder and a software decoder. On the one hand, the term "hardware encoder" refers to the CATCHY hardware system, which involves encoding complex amplitude information into a snapshot compressed measurement. The compressed 2D measurement retains the encoded complex amplitude. On the other hand, the "software decoder" represents the reconstruction algorithm used to extract the intensity and phase information from the compressed measurement.

As shown in Fig. 1, the experimental setup employs a coherent light source, namely, a He–Ne laser. Initially, the laser beam undergoes beam expansion through a 4F optical system, consisting of two achromatic lenses, AL1 (f = 30 mm) and AL2 (f = 175 mm). The light source is split into two paths by a beam splitter, BS1 (50:50). One beam passes through a mirror (M1) and then transmits through another beam splitter, BS2 (50:50), creating the reference beam $\boldsymbol{r}$. The other path reflects off the second mirror (M2) and illuminates the moving object through a microscope objective (10×, NA = 0.25), creating the object beam $\boldsymbol{o}$. The light beam carrying surface height information from the object reflects off the object's surface, then reflects off BS2, and interferes with the reference beam $\boldsymbol{r}$, forming a hologram. The object moves perpendicular to the direction of the light axis, producing dynamic holograms. The second 4F optical system composed of AL3 (f = 30 mm) and AL4 (f = 175 mm) is designed to amplify the fringe patterns of the hologram. A digital micromirror device, DMD (TI, 1920×1080 pixels, 10.8μm pixel pitch), is employed to encode dynamic holograms using different masks at different time instances, resulting in a 2D compressed measurement in the camera (MV-CA050-12UM). Mathematically, the measurement in the CATCHY system with

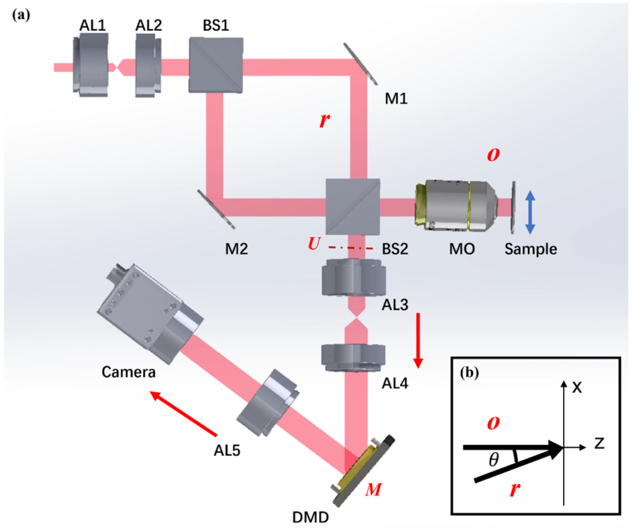


Fig. 1. (a) Our coded aperture temporal compressive digital holographic microscopy setup. AL, achromatic lens; BS, beam splitter; M, mirror; MO, microscope objective; DMD, digital micromirror device. A sample (dynamic scene) moves along the blue arrow. The red arrow indicates the light propagation direction. (b) Detail of the off-axis geometry.

T frames, each frame consists of n pixels. The process can be modeled by

$$\mathbf{y} = \mathbf{\Phi x} + \mathbf{g}, \tag{1}$$

where $\mathbf{\Phi} \in \mathbb{R}^{n \times nT}$ denotes the sensing matrix, $\mathbf{x} \in \mathbb{R}^{nT}$ is the desired signal, and $\mathbf{g} \in \mathbb{R}^{n}$ denotes the noise. The matrix $\mathbf{\Phi}$ can be written as

$$\mathbf{\Phi} = [\boldsymbol{D}_1, \ldots \boldsymbol{D}_T], \tag{2}$$

where $\{\boldsymbol{D}_t\}_{t=1}^{T}$ are diagonal matrices. Taking DHM in CACTI as an example, the hologram light field at time t is $\mathbf{U}_t$. Taking into consideration that T high-speed frames $\{\boldsymbol{U}_t\}_{t=1}^{T} \in \mathbb{R}^{n_x \times n_y}$ are subjected to modulation by the masks $\{\boldsymbol{M}_t\}_{t=1}^{T} \in \mathbb{R}^{n_x \times n_y}$. The measurement $\boldsymbol{Y} \in \mathbb{R}^{n_x \times n_y}$ is defined as

$$\mathbf{Y} = \textstyle\sum_{t=1}^{T} \mathbf{U}_t \odot \mathbf{M}_t + \boldsymbol{G}, \tag{3}$$

where $\odot$ denotes the Hadamard (element-wise) product. For every N pixel (in the t-th frame) at a given position (i, j), $i = 1, \ldots, n_x$; $j = 1, \ldots, n_y$, they are combined into one pixel in the measurement (in a single shot) as

$$y_{i,j} = \textstyle\sum_{t=1}^{T} u_{i,j,t} m_{i,j,t} + g_{i,j}. \tag{4}$$

In the first stage of the software decoding process, we adopt the video SCI reconstruction algorithm, namely, EfficientSCI [19], to obtain the holography frames from a compressed measurement. To give precise reconstruction results, we finetune the pre-trained weights of the EfficientSCI network using real masks captured by our CATCHY system. As shown in Fig. 2, in the pre-processing stage of EfficientSCI, we use the estimation module to pre-process measurements and masks, which can get a coarse estimation of the desired video $\boldsymbol{X}_p$. Then, $\boldsymbol{X}_p$ is fed into the EfficientSCI network to get the final reconstruction result. EfficientSCI is mainly composed of three parts: feature extraction module, ResDNet module, and video reconstruction module.

The feature extraction module is composed of three 3D convolutional layers. Among them, each 3D convolution is followed by a LeakyReLU activation function. The feature extraction module effectively maps the input image to the high-dimensional feature space. The ResDNet module is composed of several ResDNet blocks with dense connections within a single residual block, which can efficiently explore the spatiotemporal correlation. Specifically, the input features of the ResDNet block

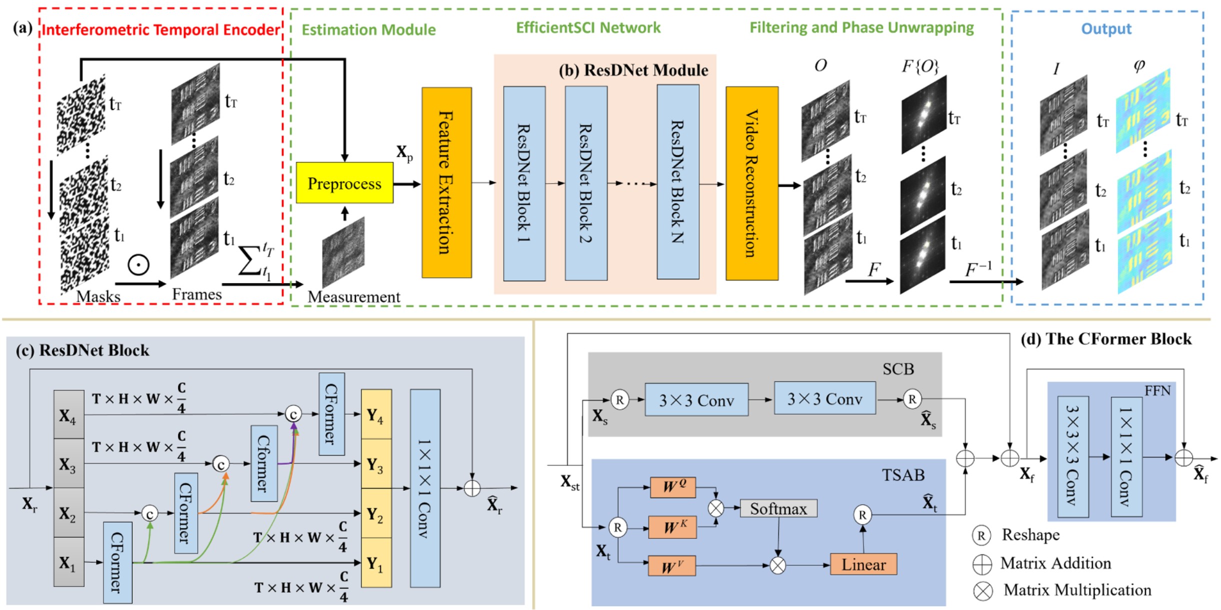


Fig. 2. (a) Illustration of the proposed reconstruction network: in the interferometric temporal encoder stage, the dynamic hologram frames from t_1 to t_T are modulated by different masks. The encoded holograms are integrated into a compressed measurement along the temporal domain, which is then fed into the EfficientSCI network. The decoded holograms are filtered by Fourier transform and inverse Fourier transform to get the first-order image, which is subjected to phase unwrapping to obtain the intensity and phase information. (b) ResDNet module consists of N ResDNet blocks. (c) Inside the ResDNet block, the input features are divided into 4 parts along the channel dimension. Each part uses CFormer to extract spatiotemporal correlation. (d) The CFormer block is composed of the spatial convolution branch (SCB), temporal self-sttention branch (TSAB), and feed forward network (FFN).

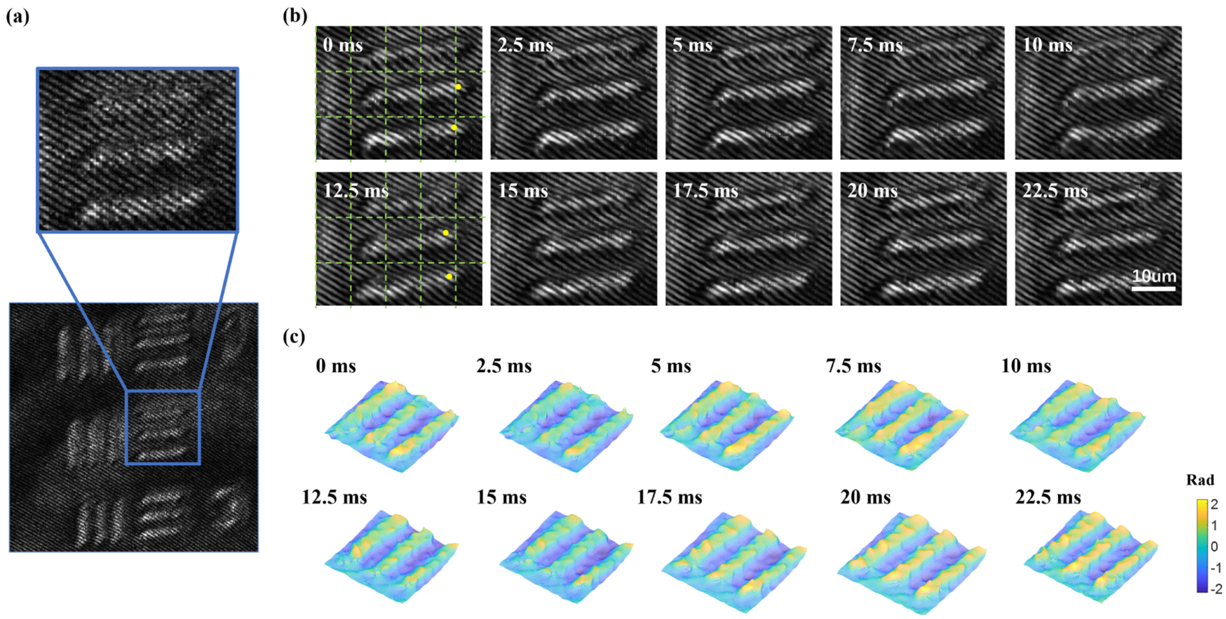


Fig. 3. CATCHY of a partial sample of 1951 USAF resolution target. (a) The 2D snapshot measurement (650 ×650 pixels) of the resolution target is compressively captured at 40 fps. Due to the compression ratio of 10, the image acquisition frame rate is equal to 400 fps. (b) Temporal holographic frames are reconstructed from a snapshot measurement. We crop 120 ×120 pixels for better visualization. See Visualization 1 for the complete video. (c) The temporal phase frames are reconstructed from holograms. During the capture, the reflective 1951 USAF resolution target moves perpendicular to the direction of the light axis. Grids and yellow dots are added to help visualize the fast-moving target.

are first divided into several parts along the feature channel dimension. Then, for each part, we use the CFormer block to efficiently establish the spatiotemporal correlation. Among them, some dense connections are added. Next, we concatenate all the CFormer output features along the feature channel dimension and adopt a 3D convolution to better fuse each part of the information. In the CFormer block, we use a spatial convolution branch (SCB) with 2D convolutions to reconstruct the local spatial details. Then, a temporal self-attention branch (TSAB) is used to calculate the temporal attention of the feature points at the same spatial position in each frame. Finally, the feed forward network (FFN) is used to further integrate the spatiotemporal information. The video reconstruction module is composed of the pixel shuffle and three 3D convolution layers to conduct video reconstruction on the output of features by the ResDNet blocks.

In the second step of the decoding process, we extract the intensity and phase information of the object from the holograms, enabling us to create a high-resolution 3D video showing the movement of the object. Initially, the holograms undergo Fourier transform processing [20]. The Fourier spectrum of the hologram is given as

$$\mathbf{f} = \mathcal{F}(|\boldsymbol{r}|^2 + |\boldsymbol{o}|^2) + \mathcal{F}(\boldsymbol{r}^*\boldsymbol{o}) + \mathcal{F}(\boldsymbol{r}\boldsymbol{o}^*), \tag{5}$$

where $\mathcal{F}(|\boldsymbol{r}|^2 + |\boldsymbol{o}|^2)$ is the zero-order spectrum, $\mathcal{F}(\boldsymbol{r}^*\boldsymbol{o})$ is the positive first-order spectrum, and $\mathcal{F}(\boldsymbol{r}\boldsymbol{o}^*)$ is the negative first-order spectrum. In off-axis holography, these three components are distinguished in the spectrum, and we choose the positive first-order term. Subsequently, by applying the inverse Fourier transform on $\mathcal{F}(\boldsymbol{r}^*\boldsymbol{o})$, the complex amplitude distribution on the recording plane can be acquired as

$$\mathbf{U} = \mathcal{F}^{-1}[\mathcal{F}(\boldsymbol{r}^*\boldsymbol{o})] = \boldsymbol{r}^*\boldsymbol{o}. \tag{6}$$

Let $\mathbf{U}_z$ denote the complex amplitude distribution on the output plane. For the reconstruction of an image-plane hologram, the propagation distance is considered as $z = 0$. The reconstruction intensity distribution on the output plane can be formulated as

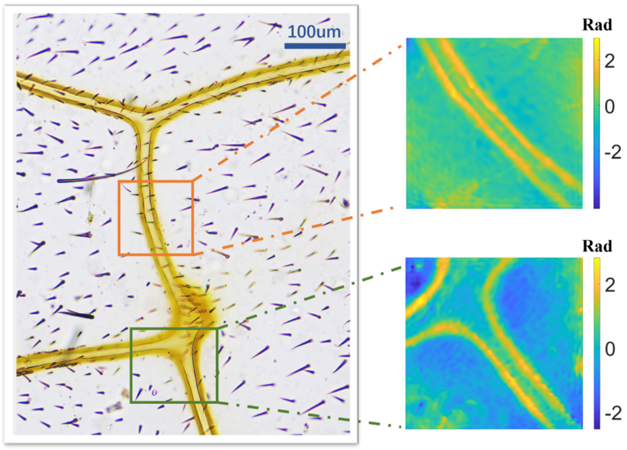
$$I_{out} = |\mathbf{U}_z|^2. \tag{7}$$

The phase distribution can be expressed as

$$\varphi = \arctan \frac{Im(\mathbf{U}_z)}{Re(\mathbf{U}_z)}, \tag{8}$$

where $Im(\mathbf{U}_z)$ is the imaginary part of $\mathbf{U}_z$. $Re(\mathbf{U}_z)$ is the real part of $\mathbf{U}_z$. Since the aforementioned φ involves wrapping and phase distortion, we employ the least square estimation for phase unwrapping and polynomial fitting for phase compensation to φ [21]. Ultimately, the effective phase for each frame can be acquired. Consequently, we can obtain 3D phase video frames.

Experimental results of our built CATCHY system are shown in Figs. 3 and 4 (see Visualization 1 and Visualization 2 for the complete videos). Two different scenes are captured by our CATCHY system. The first scene is a reflective 1951 USAF resolution target, which moves perpendicular to the direction of the light axis. The target undergoes rapid oscillations in a left-to-right motion controlled by a transnation stage. This scene aims to verify our optical system and the developed reconstruction algorithm. The captured 2D snapshot measurement of the resolution target is shown in Fig. 3(a). The 120 ×120 ×10 video frames are reconstructed from a single measurement captured within 25 ms, where the camera is running at 40 fps. With a compression ratio of 10, the acquisition frame rate reaches 400 frames per second, enabling high-speed acquisition using a low-speed camera. From Figs. 3(b) and 3(c), we can see that both the holograms and phase images are well reconstructed with sharp edges and clear motions.



(a) (b)

Fig. 4. CATCHY of a partial sample of bee wings. (a) Bee wings under a commercial microscope. (b) Single-frame phase images recovered from different measurements.

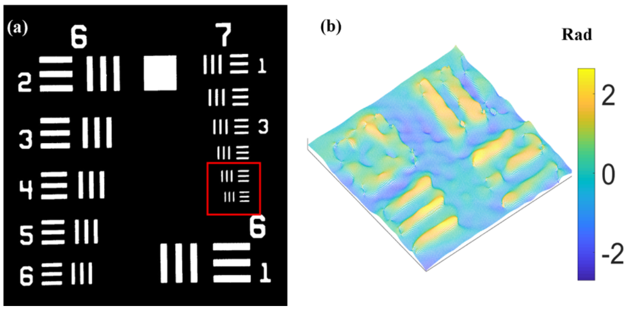


Fig. 5. (a) The elements 5 and 6 in group 7 of the resolution target. (b) Reconstructed phase image.

The second scene involves a sample of bee wings, for which we construct a Mach–Zehnder transmission-based optical setup (see Supplement 1 Fig.S1). Figure 4(a) shows bee wings observed under a commercial microscope. Figure 4(b) shows reconstructed phase images (1024 ×1024 pixels) from various bee wings' positions using CATCHY. Bee wings exhibit micron-level structures, exceeding half of the wavelength, and thus creating a wrapped region. During the recording process, the bee wings are moved using a translation stage, as shown in Visualization 2. Different from conventional microscope objective, by which we can only obtain the two-dimensional spatial position information of the bee wings, using CATCHY, we can acquire the 4D spatiotemporal information by phase imaging.

To investigate the resolution of the CATCHY system, we observe the seventh group target of standard USAF 1951 as shown in Fig. 5(a). The fast movement of the last two elements is recorded in a measurement. We record the compressive 2D measurement and observe that the sixth element of the seventh group cannot be clearly resolved, whereas the fifth element can be resolved as shown in Fig. 5 (b). Based on this observation, we determine that the resolution of CATCHY is 2.46 µm.

In this Letter, we have demonstrated the CATCHY system, which can capture high-speed high-resolution samples using the principle of snapshot compressive imaging and digital holography. A delicate reconstruction algorithm using deep learning has been developed to reconstruct the 3D scenes efficiently. Experimental results demonstrated the advantages of our system and algorithms. In the future, it will be possible to further increase the frame rate by increasing the compression ratio. Simultaneously, considering the limitation imposed by the current high resolution resulting in a small field of view, the field of view can be expanded through image stitching to address this challenge. With the help of this system, we will explore more high-speed scenarios, such as the formation of cross-scale micro-nanostructures using femtosecond laser processing, tracking liquid flow in microfluidic systems, and observing biological physiological activities. This system not only provides a means to observe high-speed and high-resolution dynamic scenes but also significantly reduces data bandwidth.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

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